Topic 6: Flexible budgeting

Sessions 15-16-17

Key topics

- Static & Flexible Budgets
- Static Budget Variances:
    - Sales / Volume Variances
    - Costs differences (variable & fixed):
    - Price (spending) Variances
    - Efficiency Variances

Static budgets vs. Flexible budgets

Static budget: budgeted revenues and costs based on the level of output **planned** (at the start of the budget period)

Flexible budget: budgeted revenues and costs based on the **actual** output (at the end of the budget period)

Variances

Variance = difference between **actual** results and **budgeted** (standard) results

Variances are used to:

- Find out and correct the problem
- Evaluate performance and motivate managers
- Potential change in the strategy

Variance Analysis

- Used to control performance
- Allows to determine the different sources of variance
- Allows management by exception

Classifying variances

Favorable: actual operating income > budgeted operating income

Unfavorable: actual operating income < budgeted operating income

Fundamentals: starting with an example

SINGADELI: bakery co. In Singapore:

- Hires 2 chefs for the baking of moon cakes. They are empowered to order raw materials and source from the best suppliers.
- They were paid 1.2 $ per cake + 50% of favourable material price variance. The co. estimates that each cake needs 50 grs/cake at a cost of 11.8 $/kgs.
- Chefs ordered 600 kgs at 9,8 $/kg.
- Total cakes: produced 11,000 ck but 2,000 ck sold at a discount of 70% (poor quality).
- Expected sales: 11.000 units at 2,25 $/cake

Fundamentals: starting with an example

CASE	
Chefs are paid:	
Per cake	1,20
Favorable material price variance	50%
Material needed (grs/cake)	50,00
Estimated cost ($/kg) by Singadeli	11,80
Orders by chefs (kg)	600,00
Cost of raw materials ($/kg)	9,80
Total cakes produced	11.000,00
Cakes of poor quality	2.000,00
Discount for poor quality	70%
Sales price per cake	2,25

Fundamentals: starting with an example

Inventory	**Budget**	**Actual**	**Variances**
Material needed (grs/cake)	50,00	50,00	0,00
Estimated cost ($/kg)	11,80	9,80	-2,00
Budgeted purchases in grs	550.000,00	550.000,00	0,00
Budgeted purchases in kg	550,00	550,00	0,00
Estimated purchases in $	6.490,00	5.390,00	-1.100,00
Real purchases in kg	550,00	600,00	50,00
Ending inventory in Kg	0,00	50,00	50,00
Cost of remaining inventory	0,00	490,00	490,00
Cost per cake	**1,79**	**1,74**	**-0,05**
Materials	0,59	0,49	-0,10
Labour fix	1,20	1,20	0,00
Labour premium	0,00	0,05	0,05

Fundamentals: starting with an example

Income Statement	Budget	Actual	Variances
Sales in units	11.000,00	11.000,00	
Price:($/cake)	2,25	2,25	
Revenues	**24.750,00**	**24.750,00**	0,00
COGS (purchases needed to produce)	6.490,00	5.390,00	-1.100,00
Gross Revenues	**18.260,00**	**19.360,00**	1.100,00
Discounts	0,00	3.150,00	3.150,00
Chef cost per cake (1,2 $)	13.200,00	13.200,00	0,00
Chef premium per savings (50%)	0,00	550,00	550,00
Net revenues	**5.060,00**	**2.460,00**	-2.600,00
ROS	28%	13%	-15%
Writing off the remaining stock	0,00	-490,00	-490,00
"Real" net revenues	**5.060,00**	**1.970,00**	-3.090,00

Fundamentals: starting with an example

- The problem arose because the chefs were paid by quantity (1,20 $/cake) + favourable cost variance, regardless of quality.
- The design of the measurement system has to be comprehensive with due consideration of all of the key factors relating to resource consumption, such as material quantity, quality, and price.
- Therefore, the management must utilize comprehensive, holistic measurement and appraisal systems that discourage the promotion of one set of metrics that may have an adverse impact on other key dimensions of business performance.
- Variance analysis enables managers to evaluate performance and learn after decisions are implemented.

Static budgets

Example assumptions (Jackets co.)

- Includes only manufacturing costs
- Units manufactured in April 2017 are sold in April 2017
- Direct materials are purchased and used in the same Budget period (no inventories)
- No work-in-process nor finished goods inventories

Data

Budget ($)	Variable cost per jacket
Direct material costs	60
Direct manufacturing labor costs	16
Variable manufacturing OH costs	12
TOTAL VARIABLE COSTS	88

Budgeted fixed costs (up to 12,000 jackets)	276.000
Budgeted selling Price ($/jacket)	120
Budgeted production and sales (jackets)	12.000
Actual production and sales (jackets)	10.000

Framework static budget

Static budget
Budgeted output
Budgeted selling Price
Budgeted variable costs/unit
Budgeted fixed costs
Operating income

Static budget (April 2017)

		Static budget
Revenues	12,000 * 120	1,440,000
Variable costs		
Direct materials	12,000 * 60	-720,000
Direct manufacturing labor	12,000 * 16	-192,000
Variable manufacturing OH	12,000 * 12	-144,000
Fixed costs		-276,000
Operating income		108,000

Static budget framework

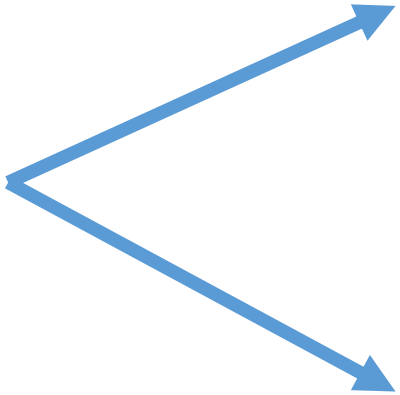
Actual results	Static budget
Actual output	Budgeted output
Actual selling Price	Budgeted selling Price
Actual variable costs/unit	Budgeted variable costs/unit
Actual fixed costs	Budgeted fixed costs
Operating income	Operating income

Static-budget variance

Actual Results (April 2017): Static B. VARIANCE

	ACTUAL	*Act. per Jkt*	*Var. per Jkt*	STATIC B.
Revenues	1.250.000	*125*	*5*	1.440.000
Variable costs	950.100			1.056.000
Direct materials	*621.600*	*62,16*	*-2,16*	*720.000*
Direct manufact. Labour	*198.000*	*19,8*	*-3,8*	*192.000*
Variable manuf. OH	*130.500*	*13,05*	*-1,05*	*144.000*
Fixed costs	285.000	*28,5*	*-5,5*	276.000
Operating income	**14.900**			**108.000**
VARIANCE (UNFAVORABLE)		**-93.100**		

Reasons for the static budget variances



Innacurate forecasting of output units

Performance in manufacturing and selling 10,000 units

Flexible budgets

Flexible budget

- ✓ Calculates budgeted revenues and costs based on the **actual output** in the Budget period.
- ✓ Calculated at the end of the budget period, once actual results are available (so retroactively prepared).
- ✓ So the Static Budget is being FLEXED or adjusted

Framework flexible budget

Flexible budget
Actual output
Budgeted selling Price
Budgeted variable costs/unit
Budgeted fixed costs
Operating income

Flexible budget

		Flexible budget
Units sold		10,000
Revenues	10,000 * 120	1,200,000
Direct materials	10,000 * 60	- 600,000
Manufac. Labor	10,000 * 16	- 160,000
Manufac. OH	10,000 * 12	- 120,000
Fixed manufacturing costs		- 276,000
Operating income		44,000

Sales/volume variance framework

Flexible budget	Static budget
Actual output	Budgeted output
Budgeted selling Price	Budgeted selling Price
Budgeted variable costs/unit	Budgeted variable costs/unit
Budgeted fixed costs	Budgeted fixed costs
Operating income	Operating income



Sales/volume variance

Sales-volume variance

Reflects the effects of *inaccurate forecasting*

of output units sold

Sales-volume variance

	Flexible budget	Sales-volume variance	Static budget
Units sold	10,000		12,000
Revenues	1,200,000		1,440,000
Direct materials	600,000		720,000
Direct manuf. Labor	160,000		192,000
Variable manuf. OH	120,000		144,000
Fixed manuf. costs	276,000		276,000
Operating income	44,000	64,000 U	108,000

Flexible-budget variance framework

Actual results	Flexible budget
Actual output	Actual output
Actual selling Price	Budgeted selling Price
Actual variable costs/unit	Budgeted variable costs/unit
Actual fixed costs	Budgeted fixed costs
Operating income	Operating income



Flexible-budget variance

Flexible-budget variance

	Actual results	Flexible-Budget Variance	Flexible budget
Units sold	10,000		10,000
Revenues	1,250,000		1,200,000
Direct materials	621,600		600,000
Direct manuf. Labor	198,000		160,000
Direct manuf. OH	130,500		120,000
Fixed manuf. Costs	285,000		276,000
Operating income	14,900	29,100 U	44,000

Static-budget variance

	Actual results	Flexible budget	Static budget
Operating income	14,900	44,000	108,000

29,100
Flexible-budget variance

64,000
Sales-volume variance

93,100
Static-budget variance

Static-budget variance framework

Actual results	Flexible budget	Static budget
Actual output	Actual output	Budgeted output
Actual selling Price	Budgeted selling Price	Budgeted selling Price
Actual variable costs	Budgeted variable costs	Budgeted variable costs
Actual fixed costs	Budgeted fixed costs	Budgeted fixed costs
Operating income	Operating income	Operating income

Flexible-budget variance **Sales/volume variance**

Static-budget variance

Direct cost inputs: variances

Flexible-budget variance: direct costs

Flexible-budget variance for direct-costs inputs is usually divided into two:

1. Price (spending) variance
2. Efficiency variance

Variable costs variances: framework

Actual results	Budgeted price	Flexible budget
Actual Output	Actual Output	Actual Output
VARIABLE COST	VARIABLE COST	VARIABLE COST



Price Variance

Efficiency Variance



Flexible-budget variance

Direct material costs variances

Direct material costs (jackets)

Budgeted cost information:

Material: $30 per sq. yard & 2 sq. yards per unit

Actual costs incurred:

Material: $28/sq. yard & 22,200 sq. yards

Production:

Budgeted: 12,000 jackets; Actual: 10,000 jackets

Direct material costs: Actual vs Budgets

	Actual	Static Budget	Flex. Budget
Material $ / sq yard	28,0	30,0	30,0
Sq yard / unit	2,22	2,00	2,00
Bud. / Actual yards used	22.200	24.000	20.000
Production	**10.000**	**12.000**	**10.000**
Bud. / Actual material costs	**621.600**	**720.000**	**600.000**

Variable costs variances: framework

Actual results	Budgeted price	Flexible budget
Actual Output	Actual Output	Actual Output
VARIABLE COST	VARIABLE COST	VARIABLE COST



Price Variance

Efficiency Variance



Flexible-budget variance

Direct materials costs budget

Actual results	Budgeted price	Flexible budget
$28 / sq. yard	$30 / sq. yard	$30 / sq. yard
22,200/10,000 = 2.22 sq yrd / unit	22,200/10,000 = 2.22 sq yrd / unit	2 sq. yard / unit
10,000 units	10,000 units	10,000 units
$621,600	$666,000	$600,000

44,400 **F**

Price Variance

66,000 **U**

Efficiency Variance

F = Favorable

U = Unfavorable

Interpreting variances

- Causes of direct materials price variances?

- Causes of direct materials efficiency variances?

Interpreting variances

- Causes of direct materials price variances?
  - ✓ Changes in price of material
  - ✓ Higher quality material
  - ✓ Higher volume of material

- Causes of direct materials efficiency variances?
  - ✓ More scrap
  - ✓ More efficiency

Direct labor costs variances

Direct labor costs

Budgeted cost information:

Labor: $20 per labor hour; 0.8 labor hours/unit

Actual costs incurred:

Labor: $22 per labor hour; 9,000 labor hours

Production:

Budgeted: 12,000 jackets; Actual: 10,000 jackets

Variable costs variances: framework

Actual results	Budgeted price	Flexible budget
Actual Output	Actual Output	Actual Output
VARIABLE COST	VARIABLE COST	VARIABLE COST

Price Variance

Efficiency Variance

Flexible-budget variance

Direct labor cost budgets

Actual results	Budgeted price	Flexible budget
$22 / LH	$20 / LH	$20 / LH
9,000 LH/10,000 units = 0.9 LH/unit	9,000 LH/10,000 units = 0.9 LH/unit	0.8 LH / unit
10,000 units	10,000 units	10,000 units
$198,000	$180,000	$160,000

18,000 U

Price Variance

20,000 U

Efficiency Variance

Interpreting variances

- Causes of direct labor price variances?
  - ✓ Increase in wages
  - ✓ Unplanned overtime
- Causes of direct labor efficiency variances?
  - ✓ More introductory workers

Variable OH cost variances

Planning manufacturing OH costs

Variable overhead costs:

- Energy
- Machine maintenance
- Engineering support
- Indirect materials

Total variable overhead budgeted: $144,000 (monthly)

Variable overhead costs

Budgeted cost information:

Total Overhead (month) = $144,000

Cost driver: Machine hours

0.4 machine hours/unit

Actual costs incurred:

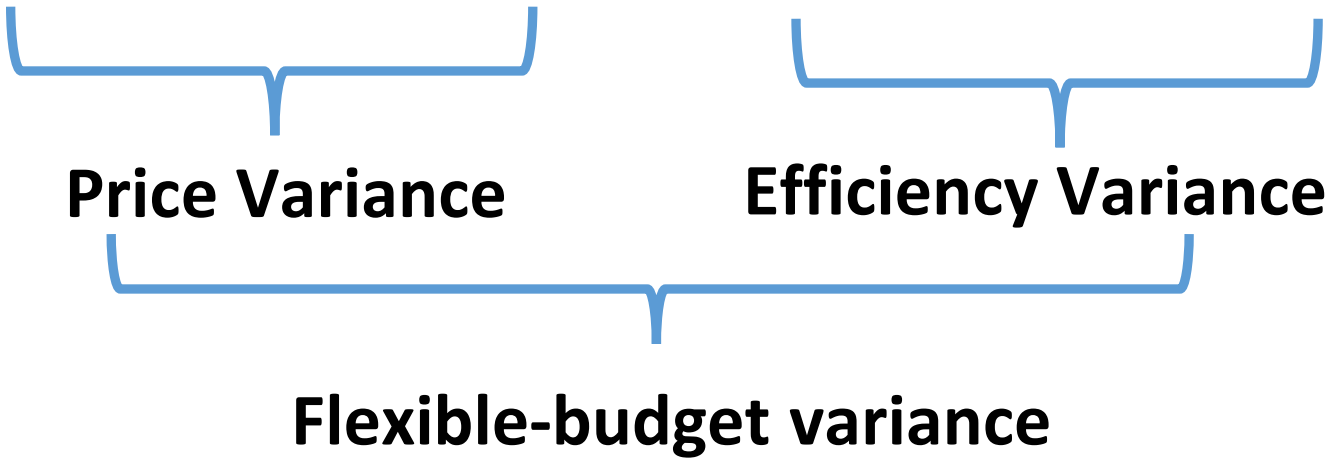
Overhead: $29/MH; 4,500 machine hours

Production:

Budgeted: 12,000 jackets; Actual: 10,000 jackets

Variable costs variances: framework

Actual results	Budgeted price	Flexible budget
Actual Output	Actual Output	Actual Output
VARIABLE COST	VARIABLE COST	VARIABLE COST



Variable overhead costs

Actual results	Budgeted price	Flexible budget
$29 / MH	$30 / MH	$30 / MH
4,500 MH/10,000 units = 0.45 MH/unit	4,500 MH/10,000 units = 0.45 MH/unit	0.4 MH / unit
10,000 units	10,000 units	10,000 units
$130,500	$135,000	$120,000

4,500 F

Price Variance

15,000 U

Efficiency Variance

Interpreting variances

- Causes of variable OH price variances?

  ✓ Prices of inputs lower than expected

  ✓ Use of energy more efficient than expected

- Causes of variable OH efficiency variances?

  ✓ Workers less skilled than planned

  ✓ Inefficient scheduling of jobs

  ✓ Bad maintenance of machines

Fixed overhead costs variances

Flexible-budget variances

	Actual results	Flexible-Budget Variance	Flexible budget
Units sold	10,000	0	10,000
Revenues	1,250,000	50,000 F	1,200,000
Direct materials	621,600	21,600 U	600,000
Direct manuf. Labor	198,000	38,000 U	160,000
Direct manuf. OH	130,500	10,500 U	120,000
Fixed manuf. Costs	285,000	9,000 U	276,000
Operating income	14,900	29,100 U	44,000

Fixed overhead costs

- Plant & equipment rent
- Plant & equipment depreciation
- Salaries of plant manager
- …

Fixed overhead costs

Budgeted cost information:

Total fixed OH costs = $276,000

Actual costs incurred:

$285,000

Production:

Budgeted: 12,000 jackets; Actual: 10,000 jackets

Fixed costs variances: framework

Actual results	Flexible budget
Actual	Budgeted
FIXED OH COSTS	FIXED OH COSTS



Price Variance

Fixed overhead costs

Actual	Flexible
Actual fixed costs	Budgeted fixed costs
285,000	276,000

9,000 U

Price Variance

Interpreting variances

- Causes of fixed OH price variances?

✓ Higher plant leasing costs

✓ Higher depreciation of plant & equipment

✓ Higher manager salaries

Take aways

- Static-buget variance can be divided into:
    1. Sales-Volume variance
    2. Flexible-Budget variance

- Difference between Static and Flexible budgets: output

- Variable cost differences between Flexible Budget and Actual Results can be divided into:
    1. Price variance
    2. Efficiency variance

References

Horngren et al., 2015: Chapters 7 & 8